INCH-POUND

MIL-C-24643/53C
14 March 1994
SUPERSEDING
MIL-C-24643/53B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, 3000 VOLTS, TYPES LSSRW, LSDRW AND LSTRW (INCLUDING VARIATIONS TYPES LSSRWA, LSCRWA AND LSTRWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: TYPES LSSRW, LSDRW AND LSTRW (unarmored) and LSSRWA, LSDRWA AND LSTRWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, AWG size 14, class B stranding.

Second - Separator (opaque white) may be used at manufacturer's option where required to provide free-stripping insulation.

* Third - Cross-linked polyethylene insulation. Insulation wall thickness 0.080 inch minimum average. Standard identification code shall be applied by method 1 or method 3. The insulation shall be colored black if method 1 is used.

Fourth - The required two or three conductors cabled together with a lay not greater than 24 times the pitch diameter. Fillers shall be used as necessary to form a firm, well-rounded assembly.

Fifth - Binder tape applied helically with overlap for two and three conductor cables only.

Construction variant for type LSSRW, LSDRW AND LSTRW

* Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.

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Construction (watertight) (continued)

Construction variant for type LSSRWA, LSDRWA AND LSTRWA

* Sixth - Cross-linked polyolefin jacket. Cable jacket thickness 0.050 inch minimum average.

* Seventh- Braided metal armor.

TABLE I. Details

Military 1/ part no. M24643/53	Cable type	Number of conductors	Overall diameter maximum (inch)	Conductor resistance (ohms)	Cold bending mandrel diameter (max inch)
-01UN	LSSRW	1	0.400	2.69	3
-02UN	LSDRW	2	.670	2.75	5
-03UN	LSTRW	3	.710	2.75	7
-04UN	LSSRWA	1	.450	2.69	3
-05UN	LSDRWA	2	.720	2.75	5
-06UN	LSTRWA	3	.760	2.75	7

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.

EXAMINATION AND TESTS:

Basic electrical:	Requirements
Conductor Resistance - ohms/1000 feet at 25°C, maximum	(see table I)
Voltage withstand - volts, root mean square, minimum	8000
* Insulation resistance - Conductor to conductor, Megohms/1000 feet, minimum	9500
Conductor continuity	No failure
Group A:	
Visual and dimensional	No failure
Watertightness - see MIL-C-24643 for water leakage	No failure
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Cold bending, cable - (see table I for mandrel diameter)	No damage
Drip - 95 ±1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250

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EXAMINATION AND TESTS (continued):

	Requirements
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160
Group C:	
Physicals - (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
* Elongation - percent of unaged, minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (conductor) - (method 1 only) cycles, minimum (LSDRW and LSTRW)	25
* Permanence of printing (jacket) - (type LSSRW, LSDRW and LSTRW) cycles, minimum	125
Cable sealant removability	No failure
* Armor - conformance to material, construction and coverage (type LSSRWA, LSDRWA and LSTRWA only)	No failure
Group D:	
* Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ±5°C.)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2

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QUALIFICATION INSPECTION (continued):

	Requirements
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS:

Type and size	Feet (nominal)
LSSRW	2500
LSDRW and LSTRW	2000

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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